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Text-to-SQL Evaluation Toolkit

A modular, open-source framework for rigorous and reproducible evaluation of text-to-SQL systems.

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12+	6	4	2	Apache 2.0
EVALUATION METRICS	PRE-LOADED BENCHMARKS	DATABASE ENGINES	INFERENCE PIPELINES	ON PYPI & GITHUB

The Evaluation Gap

LLMs made text-to-SQL generation strong. **Measuring** it did not keep up.

1. **SQL semantics are subtle**
   Column ordering, aliasing and `DISTINCT` may be equivalent — or divergent. String matching cannot tell.
2. **Execution metrics disagree**
   Implementations differ on column sets, row ordering and empty results, so published scores are not comparable.
3. **Many queries are correct**
   One question admits many valid formulations, yet most frameworks allow a single ground truth.
4. **No shared implementation**
   Benchmark-specific ad hoc scripts are hard to reproduce, extend or compare.

What the Toolkit Provides

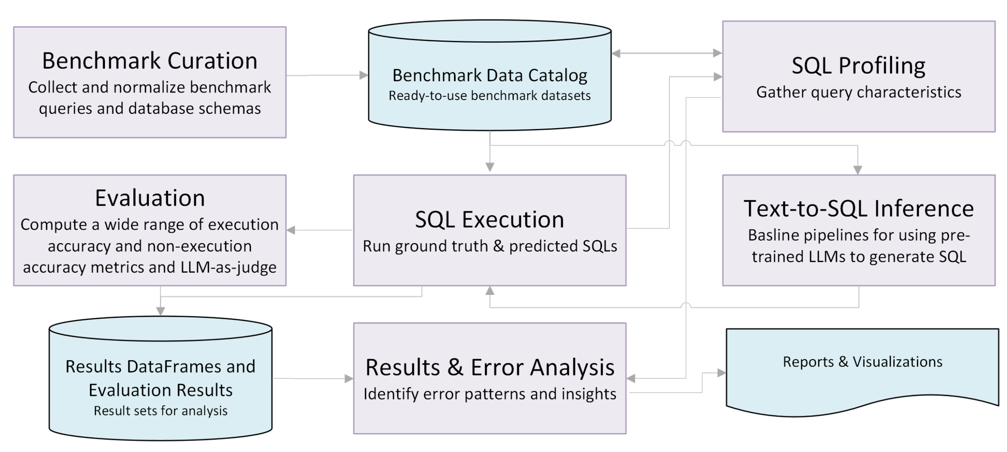
- **Evaluation engine.** Twelve-plus metrics across execution accuracy, SQL syntactic equivalence, parsability and LLM-as-judge, with multiple ground truths.
- **Integrated pipelines.** Standard and agentic LLM inference, execution against real databases, SQL profiling, error analysis.
- **Interactive dashboard.** Browse benchmarks, compare pipelines, analyse errors and re-evaluate single records live.
- **Reproducible release.** Apache 2.0, on PyPI, validated on established public benchmarks — a common foundation for the community.

Beyond a Single Number

- **Relaxed variants.** Non-empty and subset non-empty accuracy drop trivially correct empty matches and forgive ambiguous column requirements: "list customers" may validly return IDs, names or both.
- **Multiple ground truths.** A question may carry several gold queries; the maximum is reported, so transposed rows, alternative joins and rounding stop penalising correct predictions.
- **Judge where metrics stall.** An inconclusive or missing ground truth triggers an LLM judge over the question, both queries and both result frames — 1/0.5/0 with an explanation, skipped when subset EX already agrees.

Modular Architecture

Components share one JSON prediction format, so a team can adopt a single component or the whole pipeline.



> Adding a benchmark is **one JSON entry** — questions, schema, prediction files, database connection. No code changes.

Engines: SQLite • PostgreSQL • MySQL • Db2.
Benchmarks: BIRD Mini-Dev • Spider Dev • Spider Realistic • Beaver • Archer.

Evaluation Metrics

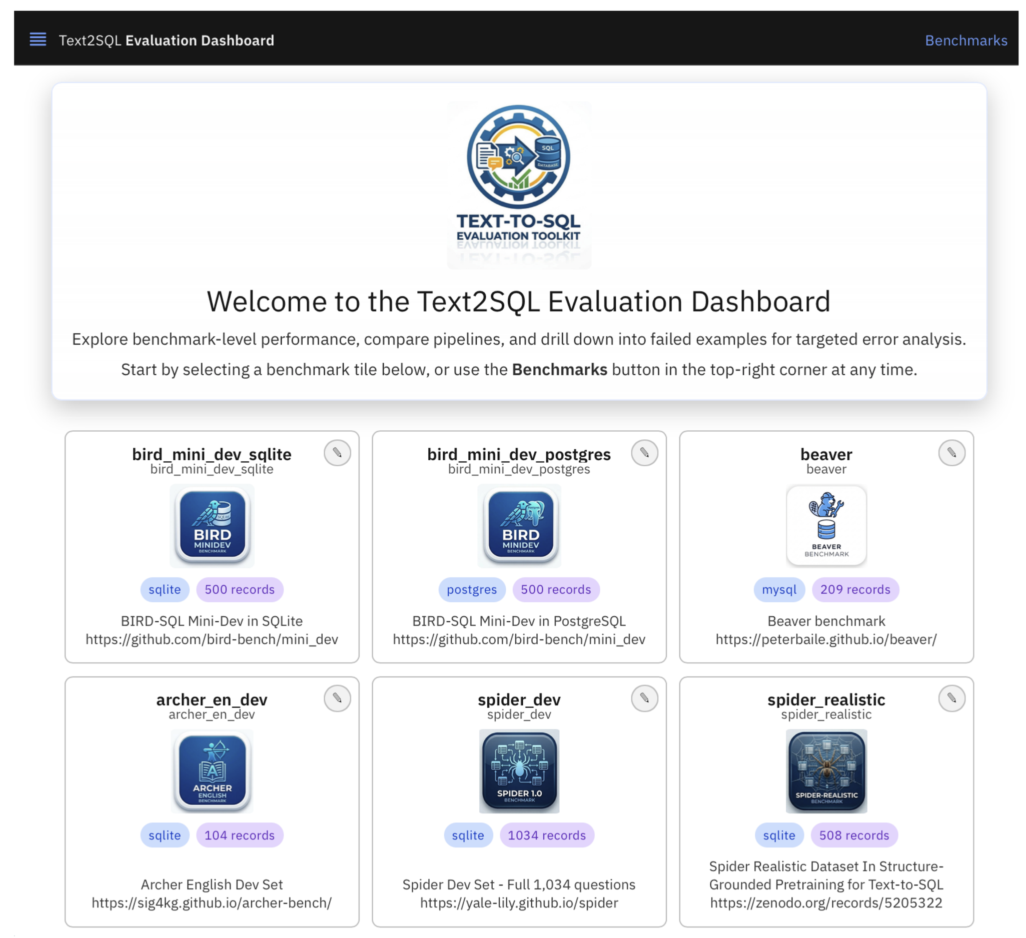
Metric	Description
EXECUTION-BASED	
Execution Accuracy (EX)	Exact result-set match
Non-Empty EX	Both results non-empty
Subset Non-Empty EX	Allows extra/missing columns
BIRD EX	Set-based match, BIRD rules
SQL SYNTACTIC EQUIVALENCE	
SQL Exact Match (EM)	Exact string match
SQLGlot EM	AST equivalence via SQLGlot
SQLGlot Optimized EM	Equivalence after optimization
SQLParse EM	Token-based equivalence
Syntactic EM	Any of the above holds
PARSABILITY	
SQLGlot Parsable	Prediction parses with SQLGlot
SQLParse Parsable	Prediction parses with sqlparse
LLM-BASED	
LLM-as-Judge Score	Correctness verdict (0 / 0.5 / 1)
EFFICIENCY	
Inference Time	LLM inference latency (ms)
Execution Time	SQL execution time (ms)
Token Usage	Prompt, completion, total tokens

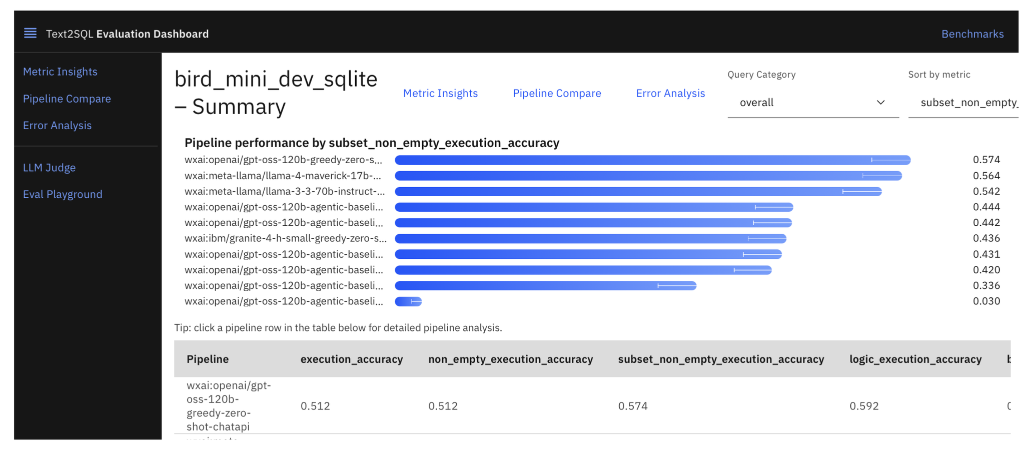
Evaluate in Three Lines

```python
from text2sql_eval_toolkit import run_evaluation
data, summary = run_evaluation("bird_mini_dev_sqlite")
print(summary)  # per-pipeline metric averages
```

Any system can be scored by supplying predictions as JSON — question id, predicted SQL, optional metadata. Adopting our inference pipeline is optional.

Interactive Dashboard





- **Benchmark overview.** Tiles per benchmark; per-pipeline metrics stratified by query difficulty.
- **Pipeline comparison.** Cross-pipeline confusion matrices with timing and token deltas; click a cell to drill into the records.
- **Error analysis.** Full-text search, metric-condition filters (`EX=0 & llm_score=1`) and cross-pipeline disagreement.
- **Playground.** Write new SQL, re-execute and re-score live; edit the judge prompt in YAML. A REST API exposes the same workflows to CI/CD.

Demonstration Walkthrough

1. **Benchmark exploration**
   Open BIRD Mini-Dev from six pre-loaded benchmarks, then contrast a public benchmark with an enterprise one.
2. **Errors and comparison**
   Zero-shot baseline versus an agentic self-correcting pipeline: recoveries, regressions, and the cost/accuracy trade-off.
3. **Metrics vs. LLM-as-judge**
   Find predictions the strict metric scores 0 and the judge scores 1 — and the cases where the judge is the one that is wrong.
4. **Interactive evaluation**
   Write your own SQL for any question and watch the full metric suite score it live, judge toggled on and off.

Code & docs
github.com/IBM/text2sql-eval-toolkit

Demo video
youtu.be/PkKrOmazqmI

`pip install text2sql-eval-toolkit`
Apache 2.0 • contributions welcome
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